



Crash Course in Maths for Cognitive Science

Session 3 | Linear Algebra I

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What today is about

Linear algebra is the mathematics of *data in many dimensions* and of *linear transformations*. It is the language of almost every model you will meet later.

- A *vector*: a point, an arrow, or a list of features.
- The *dot product*: similarity, length, angle, and projection.
- A *matrix*, read two ways: a data table *and* a transformation.
- *Linear systems*, the *inverse*, and *rank*, the true dimensionality of data.

Everything is concrete: we will draw almost every idea as a picture, and solve several small examples on each concept.

The single operation everything is built from

Linear combination

Take some vectors, scale each by a number, and add them up:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k.$$

Almost every concept today (span, matrix multiplication, and next session's PCA) is really a statement about linear combinations.

If this operation feels natural by the end of the session, everything else follows.

A vector, read three ways

Definition

A **vector** $v \in \mathbb{R}^n$ is an ordered list of numbers. We can read it as a *point* at those coordinates, an *arrow* from the origin to that point, or a *list of features* describing one object.

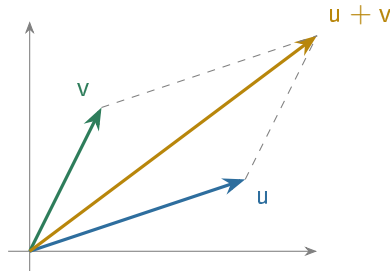
All three readings matter. “Arrow” gives us geometry (length, direction, angle). “Point” lets us picture a whole dataset as a cloud of points. “List of features” is how real data arrives: a participant is a vector of (age, reaction time, accuracy, ...); an image is a vector of pixel values.

Adding and scaling vectors

We combine vectors in just two ways:

- **Add** them componentwise, geometrically, place them tip-to-tail.
- **Scale** them by a number, stretch the arrow, or flip it if the number is negative.

Every richer operation is built from these two.



Linear combination: try it

Example

Let $u = (3, 1)$ and $v = (1, 2)$. Compute the linear combination

$$w = 2u - v,$$

and describe where its arrow points.

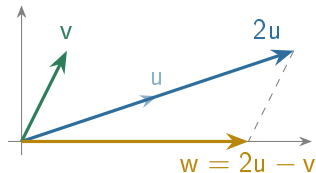
Try it yourself first — the solution is on the next slide.

Linear combination: solution

Scale first, then add componentwise:

$$2u = (6, 2), \quad w = (6, 2) - (1, 2) = (5, 0).$$

Doubling u stretches its arrow; subtracting v pulls it back down onto the horizontal axis. Every point you can build this way is a *linear combination* of u and v , and with these two directions you can reach *any* point in the plane.



The dot product: one operation, three meanings

Definition

The **dot product** multiplies matching components and adds:

$$u \cdot v = \sum_i u_i v_i = \|u\| \|v\| \cos \theta,$$

where θ is the angle between the arrows.

From it we also read off *length* $\|v\| = \sqrt{v \cdot v}$ and *distance* $\|u - v\|$. So a single number quietly encodes similarity, size, and angle all at once.

The dot product is a similarity score

Intuition

Vectors pointing the same way have a large positive dot product; perpendicular vectors have dot product zero (“nothing in common”); opposite vectors have a large negative one.

Divide the dot product by the two lengths and you get $\cos \theta$, the **cosine similarity** used everywhere to compare feature vectors, because it ignores their overall magnitude and keeps only their direction.

Dot product and angle: try it

Example

Let $u = (1, 2)$ and $v = (3, 1)$. Find $u \cdot v$, the two lengths, and the angle θ between the arrows.

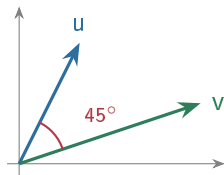
Try it yourself first — the solution is on the next slide.

Dot product and angle: solution

The dot product is $u \cdot v = 1 \cdot 3 + 2 \cdot 1 = 5$, and the lengths are $\|u\| = \sqrt{5}$, $\|v\| = \sqrt{10}$. So

$$\cos \theta = \frac{5}{\sqrt{5} \sqrt{10}} = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ.$$

A positive dot product means an acute angle, a moderate similarity in direction.



Cosine similarity: try it

Example

Two neural population responses to two stimuli are recorded on three cells:

$$a = (1, 2, 2), \quad b = (2, 2, 1).$$

How similar are the two response *patterns*? Use cosine similarity, which ignores overall firing level and keeps only the pattern.

Try it yourself first — the solution is on the next slide.

Cosine similarity: solution

The dot product is $a \cdot b = 2 + 4 + 2 = 8$, and both vectors have length $\|a\| = \|b\| = \sqrt{1 + 4 + 4} = 3$. So

$$\cos \theta = \frac{8}{3 \cdot 3} = \frac{8}{9} \approx 0.89 \quad \Rightarrow \quad \theta \approx 27^\circ.$$

Intuition

A cosine near 1 means the two patterns rise and fall together, the population “says almost the same thing” about the two stimuli, even if one response is overall stronger. Cosine near 0 would mean unrelated patterns; near -1 , opposite ones.

A few facts about the dot product

Useful fact

Cauchy–Schwarz: $|u \cdot v| \leq \|u\| \|v\|$, which is why cosine similarity always lies in $[-1, 1]$.

Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$, a detour is never shorter than the direct route.

Unit vector: $\hat{u} = u / \|u\|$ keeps only the direction.

Orthogonality: the geometry of “unrelated”

Key idea

Two vectors are *perpendicular (orthogonal)* exactly when their dot product is zero. This is the geometric meaning of “no linear relationship”, and it is the same idea as zero correlation once data are mean-centred.

It connects directly to last session’s regression: the best-fit line is the one whose residual vector is orthogonal to the predictors. That is the bridge we make concrete next.

Projection: the geometry of least squares

Useful fact

The projection of b onto the direction of a is

$$\text{proj}_a b = \frac{a \cdot b}{a \cdot a} a,$$

and the leftover (the *residual*) is orthogonal to a .

The projection is the “shadow” of b along the direction of a ; the residual is the part of b that a simply cannot explain. Fitting a regression line is exactly projecting the data onto the space of predictors.

Projection: try it

Example

Project $b = (2, 3)$ onto the direction of $a = (1, 1)$. What is the shadow, and what is the leftover residual? Check that the residual is really perpendicular to a .

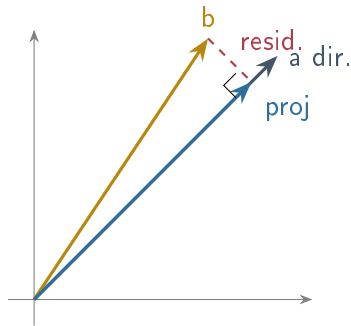
Try it yourself first — the solution is on the next slide.

Projection: solution

Apply the formula:

$$\text{proj}_a b = \frac{a \cdot b}{a \cdot a} a = \frac{5}{2}(1, 1) = (2.5, 2.5).$$

The residual is $b - \text{proj}_a b = (-0.5, 0.5)$, and indeed $a \cdot (-0.5, 0.5) = -0.5 + 0.5 = 0$, perpendicular, exactly as the lemma promises.



Link to cognitive science

Representing stimuli, words, or neural population states as vectors, and comparing them by dot product or cosine similarity, is the backbone of *representational similarity analysis* in neuroscience and of *word embeddings* in computational linguistics.

“King – man + woman $\approx$ queen” is literally vector arithmetic. A population of neurons firing is a single point in a high-dimensional space, and how two conditions differ is just a distance between two such points.

Practice: vectors and the dot product

Practice — try it

Let $u = (3, 4)$ and $v = (4, -3)$.

1. Find $\|u\|$.
2. Compute $u \cdot v$. Are the two vectors orthogonal?
3. Give the unit vector in the direction of u .

Answers. (1) $\|u\| = 5$. (2) $u \cdot v = 0$, so yes, orthogonal. (3) $(0.6, 0.8)$.

A matrix has two readings: keep both

Definition

An $m \times n$ **matrix** is a rectangular grid of numbers. Read it as a *data table* (rows are observations, columns are features) *and* as a *linear transformation* $x \mapsto Ax$ that moves every vector to a new place.

The data-table reading is how we *store* observations. The transformation reading is how models *act* on them, for example, a layer of a neural network multiplies its input by a weight matrix.

A matrix transforms space

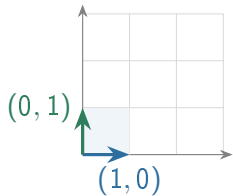
Intuition

A matrix sends the whole grid of vectors to a new grid, keeping the origin fixed and straight lines straight. The *columns* tell you exactly where the basis vectors land: the first column is where $(1, 0)$ goes, the second where $(0, 1)$ goes. Everything else follows by linear combination.

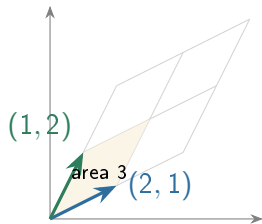
Rotations, reflections, scalings, and shears are all just different matrices, different instructions for where the two axis arrows should end up.

Seeing a matrix act on the whole grid

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ sends $(1, 0) \rightarrow (2, 1)$ and $(0, 1) \rightarrow (1, 2)$, its two columns. The unit square becomes the parallelogram they span.



Before: the unit square, area 1.



After: the unit cell becomes a parallelogram, area $\det A = 3$.

Matrix times vector = combination of columns

The product Ax is a linear combination of the columns of A , weighted by the entries of x :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} a \\ c \end{pmatrix} + y \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

Intuition

Reading a matrix–vector product as “mix the columns” is the single most useful habit in linear algebra, it turns $Ax = b$ into “which mix of the columns makes b ?”

Matrix $\times$ vector: try it

Example

Let $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ and $x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Compute Ax two ways: row-by-row, *and* as a combination of the columns of A .

Try it yourself first — the solution is on the next slide.

Matrix $\times$ vector: solution

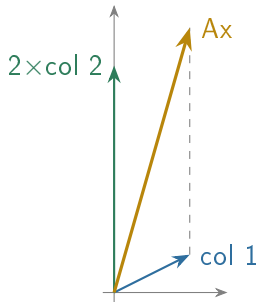
Row by row:

$$Ax = \begin{pmatrix} 2 \cdot 1 + 0 \cdot 2 \\ 1 \cdot 1 + 3 \cdot 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}.$$

As a column mix:

$$1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}.$$

Same answer, two readings of one operation.



A matrix as a specific transformation

A rotation

$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ sends $(1, 0)$ to $(0, 1)$ and $(0, 1)$ to $(-1, 0)$: the arrow pointing right becomes the arrow pointing up, a quarter turn anticlockwise. The columns of R are exactly where the two axis vectors land.

This is the payoff of “columns say where the basis goes”: to read what a matrix *does*, just look at its columns.

Multiplying two matrices = composing transformations

The product AB means “do B , then A ”. Entry (i,j) is the dot product of row i of A with column j of B .

Intuition

Matrix multiplication is just “apply one transformation after another”. That is why the inner dimensions must match, and why order matters, doing B then A is generally not the same as A then B .

Example

Compute the product

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The right factor swaps a standard pair of things, which? Predict the effect before you compute.

Try it yourself first — the solution is on the next slide.

Each entry is a row-times-column dot product:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 \cdot 0 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 0 \\ 3 \cdot 0 + 4 \cdot 1 & 3 \cdot 1 + 4 \cdot 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

The matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is the “swap” map; right-multiplying by it **swapped the columns** of the first matrix. (Left-multiplying by it would swap the *rows* instead, order matters.)

The algebra of matrices: and one warning

Useful fact

Matrix multiplication is *associative* and *distributive*; $(AB)^T = B^T A^T$; and $AI = A$.

Common pitfall

Matrix multiplication is **not commutative**: in general $AB \neq BA$. Order encodes the order of operations, “rotate then stretch” differs from “stretch then rotate”. Also, inner dimensions must match: an $m \times n$ times an $n \times p$ gives $m \times p$.

Order matters: see it for yourself

Example

With $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ (two different shears), compute both AB and BA . Are they equal?

Try it yourself first — the solution is on the next slide.

Order matters: solution

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Common pitfall

$AB \neq BA$: the two orders give genuinely different matrices. “Shear horizontally then vertically” is not the same transformation as the reverse. Always keep track of which side you multiply on.

Practice: matrices

Practice — try it

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ (a horizontal shear) and $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

1. Compute Ax .
2. Compute AA (apply the shear twice).
3. Does A change area? Find $\det A$ (we build this next).

Answers. (1) $Ax = (3, 1)$. (2) $AA = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$. (3) $\det A = 1$, area unchanged.

A system of equations is one matrix equation

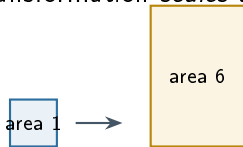
Definition

The system $\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$ is written compactly as $Ax = b$. Solving it means finding the vector x that A maps to b .

Two ways to picture it. *Row picture*: each equation is a line, and the solution is where the lines cross. *Column picture*: which combination of A 's columns produces b ?

A gentle detour: how much does a matrix stretch area?

Before we can “undo” a matrix, one number tells us a lot: the **determinant**. It measures how much the transformation *scales area*.



$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ stretches $\times 2$ across and $\times 3$ up: area $\times 6$.



A *shear* $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ slides the top over but keeps the area: still 1.

Computing the determinant of a 2×2

Definition

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is

$$\det A = ad - bc.$$

Its size $|\det A|$ is the area-scaling factor.

Example

$\det\left(\begin{smallmatrix} 3 & 1 \\ 2 & 4 \end{smallmatrix}\right) = 3 \cdot 4 - 1 \cdot 2 = 12 - 2 = 10$. This matrix multiplies areas by 10.

Determinant and invertibility: try it

Example

Compute both determinants and say which matrix can be undone:

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}.$$

Try it yourself first — the solution is on the next slide.

Determinant and invertibility: solution

$$\det A = 2 \cdot 4 - 3 \cdot 1 = 8 - 3 = 5 \neq 0, \quad \det B = 4 \cdot 1 - 2 \cdot 2 = 4 - 4 = 0.$$

Intuition

A scales area by 5: nothing is lost, so it can be undone. B has $\det = 0$: its two columns $(4, 2)$ and $(2, 1)$ lie on one line, so it crushes the plane onto that line and *cannot* be undone. A zero determinant is the exact warning sign of non-invertibility.

When the determinant is zero, the matrix collapses space

Intuition

If $\det A = 0$, the area-scaling factor is 0: the transformation squashes the whole plane onto a single *line* (or a point). Different inputs get mapped to the same output, so you can no longer tell them apart; the matrix cannot be undone.



$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: the two columns lie on one line, so the square is crushed onto it. $\det = 0$.

A zero determinant is the warning sign that a system may have *no* solution or *infinitely many*. A nonzero determinant guarantees exactly one.

Undoing a transformation: the inverse

Definition

The **inverse** A^{-1} undoes A : $A^{-1}A = I$. It exists exactly when $\det A \neq 0$ (the matrix didn't collapse anything). For a 2×2 matrix,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and then $Ax = b$ has the unique solution $x = A^{-1}b$.

Inverse: try it

Example

Find the inverse of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, then check your answer by multiplying AA^{-1} .

Try it yourself first — the solution is on the next slide.

Inverse: solution

The determinant is $\det A = 1 \cdot 4 - 2 \cdot 3 = -2 \neq 0$, so the inverse exists:

$$A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}.$$

$$\text{Check: } \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix} = \begin{pmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I. \checkmark$$

Solving a system with the inverse: try it

Example

Use the inverse you just found to solve $Ax = b$ with $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $b = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$. Then check by plugging back in.

Try it yourself first — the solution is on the next slide.

Solving a system with the inverse: solution

Since $x = A^{-1}b$,

$$x = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -10 + 6 \\ 7.5 - 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 4.5 \end{pmatrix}.$$

Check: $Ax = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -4 \\ 4.5 \end{pmatrix} = \begin{pmatrix} -4 + 9 \\ -12 + 18 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix} = b$. ✓ The inverse turns solving a system into a single multiplication.

Solving larger systems by elimination

Gaussian elimination: use one equation to cancel a variable from the others, repeat, then back-substitute. It is the 2×2 idea, scaled up, and it is what software actually does, since forming A^{-1} is wasteful for large systems.

Example

Solve

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

Try it yourself first — the solution is on the next slide.

Gaussian elimination: solution

Subtract multiples of the first equation to remove x from the other two:

$$(2x - y + z) - 2(x + y + z) \Rightarrow -3y - z = -9, \quad (x + 2y - z) - (x + y + z) \Rightarrow y - 2z = -4.$$

From $y - 2z = -4$ we get $y = 2z - 4$; substituting into $3y + z = 9$ gives $7z = 21$, so $z = 3$, then $y = 2$ and finally $x = 6 - y - z = 1$. The solution is

$$(x, y, z) = (1, 2, 3).$$

Back-check: $1 + 2 + 3 = 6$, $2 - 2 + 3 = 3$, $1 + 4 - 3 = 2$. ✓

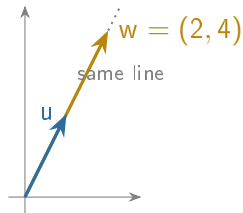
When is a vector “redundant”?

Look at $u = (1, 2)$ and $w = (2, 4)$.

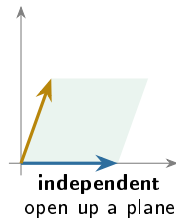
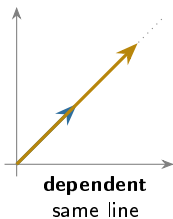
w is just $2 \times u$, it points in the *same* direction.

Once you have u , adding w tells you nothing new:
both lie on one line.

We say u and w are **linearly dependent**: one is a combination of the other.



Independent vs. dependent: the picture



Key idea

Vectors are **independent** when none is a combination of the others, each points in a genuinely new direction.

Span: what can you reach?

Definition

The **span** of some vectors is everything you can reach by linear combinations of them (scaling and adding).

- Two vectors on the *same line* span only that *line*.
- Two vectors in *different directions* span the whole *plane*, you can reach any point by mixing them.

So “how many independent directions you have” decides “how big a space you can reach”.

Rank = the number of independent directions

Key idea

The **rank** of a matrix is how many of its columns point in genuinely different directions, the number of independent directions in the data. It is the *honest* count of dimensions, which can be far fewer than the number of columns.

Two vectors on the same line: rank 1. Two vectors opening a plane: rank 2. The rank never exceeds the number of columns, or the number of rows.

Example

Find the rank of each matrix by spotting which columns are genuinely new directions:

$$M = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \end{pmatrix}, \quad N = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}.$$

Try it yourself first — the solution is on the next slide.

Rank: solution

M: column 2 = $(3, 6) = 3 \times$ column 1 (no new direction), but column 3 = $(2, 5)$ is *not* a multiple of column 1, a genuinely new direction. Two independent directions, so rank M = 2.

N: every column is a multiple of $(1, 2)$: $(2, 4) = 2 \times (1, 2)$ and $(3, 6) = 3 \times (1, 2)$. All three lie on one line, just one independent direction, so rank N = 1.

Intuition

Rank counts *directions*, not columns. Three columns can still hide only one real direction.

Quick tests for independence

Useful fact

- Two vectors are dependent exactly when one is a multiple of the other.
- For a square matrix, the columns are independent exactly when $\det \neq 0$.
- Any more than n vectors in $\mathbb{R}^n$ are automatically dependent (you cannot have three independent directions in a plane).

Why rank matters in neuroscience

Link to cognitive science

This is exactly what we mean by the *dimensionality* of neural data. A recording of 100 neurons lives in $\mathbb{R}^{100}$, but their activity during a task often varies along only a handful of independent directions, the data have low rank.

Finding those few directions is *dimensionality reduction*, and the principled way to do it is Principal Component Analysis, exactly where next session begins.

Practice: rank and independence

Practice — try it

1. Are $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ independent?
2. Are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ independent?
3. Can three vectors in $\mathbb{R}^2$ ever be independent?

Answers. (1) dependent (one is $2\times$ the other). (2) independent. (3) no, never in $\mathbb{R}^2$.

Session 3: what we built

- Vectors are points/arrows/features, combined by *linear combination*.
- The *dot product* gives length, distance, angle, and similarity; *orthogonality* is “unrelated”; *projection* is least squares.
- A *matrix* is both a data table and a transformation; the *determinant* is its area-scaling factor.
- The *inverse* exists when $\det \neq 0$; *rank* counts the independent directions, the true dimensionality of data.

Next: the special directions of a matrix, eigenvectors, and their flagship application, PCA.